



# 8846 - The Polar Expedition: Extracting Spectra of Polar Regions in Planetary-mass Objects and Brown Dwarfs

Cycle: 4, Proposal Category: GO

## INVESTIGATORS

| <i>Name</i>                                   | <i>Institution</i>           |
|-----------------------------------------------|------------------------------|
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## OBSERVATIONS

| <i>Folder</i> | <i>Observation</i> | <i>Label</i>                                   | <i>Observing Template</i>         | <i>Science Target</i> |
|---------------|--------------------|------------------------------------------------|-----------------------------------|-----------------------|
| EPOCH-A       |                    |                                                |                                   |                       |
|               | 1                  | 1A) 2M2139, ngroups=50, int/expo=303           | NIRSpec Bright Object Time Series | (1) 2M2139+02         |
|               | 2                  | 2A) PSO 318, ngroups=100x2, exposures/epoch=34 | NIRSpec Bright Object Time Series | (2) PSO318            |
|               | 3                  | 3A) 2M1126, ngroups=83, int/expo=76            | NIRSpec Bright Object Time Series | (3) 2M1126-50         |
| EPOCH-B       |                    |                                                |                                   |                       |
|               | 4                  | 1B) 2M2139, ngroups=50, int/expo=303           | NIRSpec Bright Object Time Series | (1) 2M2139+02         |
|               | 5                  | 2B) PSO 318, ngroups=100x2, exposures/epoch=34 | NIRSpec Bright Object Time Series | (2) PSO318            |
|               | 6                  | 3B) 2M1126, ngroups=83, int/expo=76            | NIRSpec Bright Object Time Series | (3) 2M1126-50         |

| <i>Folder</i> | <i>Observation</i> | <i>Label</i>                                   | <i>Observing Template</i>         | <i>Science Target</i> |
|---------------|--------------------|------------------------------------------------|-----------------------------------|-----------------------|
| EPOCH-A       |                    |                                                |                                   |                       |
|               | 7                  | 1C) 2M2139, ngroups=50, int/expo=303           | NIRSpec Bright Object Time Series | (1) 2M2139+02         |
|               | 8                  | 2C) PSO 318, ngroups=100x2, exposures/epoch=34 | NIRSpec Bright Object Time Series | (2) PSO318            |
|               | 9                  | 3C) 2M1126, ngroups=83, int/expo=76            | NIRSpec Bright Object Time Series | (3) 2M1126-50         |

## ABSTRACT

We propose a unique, time-resolved, multi-epoch program for three L-T transition ultracool dwarfs and planetary-mass objects, viewed from pole-on to equator-on orientations to test the polar-vortex hypothesis. The polar vortex hypothesis explains observed color--inclination and spectrophotometric trends by assuming that polar regions are dynamically and spectrally distinct from the equatorial and mid-latitude regions. If the polar vortex hypothesis is correct, we will isolate two distinct sets of spectra: one from the polar and one from the equatorial regions. With our JWST program, we will be able to test the existence of polar vortices outside the Solar System for the first time, and tackle an outstanding question: Are there spectrally distinct polar regions of different color, cloud properties, and dynamical timescales in brown dwarfs and planetary-mass objects — like in Jupiter?

## OBSERVING DESCRIPTION

We propose for NIRSpec G235H/F170LP Bright Object Time-Series (BOTS) observations of three targets: 2M2139+02, PSO 318, and 2M1126-50 to test the polar vortex hypothesis. To trace spectrophotometric evolution over multiple long- and short-timescales, we propose to probe 6 full rotations over 3 epochs for each object, with 2 rotations per epoch, and a minimum separation of 15x rotations between epochs.

The G235H/F170LP high spectral resolution from 1.66-3.01 $\mu$ m would allow the best-quality spectroscopy of our targets, specifically providing a better probing of pressure levels in the contribution function; better constraint on clouds slopes from condensate particles up to radii 3-5 $\mu$ m; better probes for H<sub>2</sub>O, CH<sub>4</sub>, Na I and K I; crucial access to CO lines to infer a P-T profile location and identify cloud-decks position; and better constraint of  $T_{\text{eff}}$ .

We require photometric precision of at least 1/5th of the object's J-band variability percentage to guarantee stable spectrophotometric monitoring. Because our targets have different J-band variability amplitudes, we aim for different flux-change precision of 3%, 1%, and 0.25%, corresponding to

SNR of 50, 100, and 400 for 2M2139, PSO 318, and 2M1126 respectively.

We performed the exposure time calculation (ETC: 224186) following the recommendations of the ETC guide. We used the ETC-included  $T_{\text{eff}}=1200$  K BTSettl brown dwarfs model spectra for all our targets as they are relatively similar in effective temperatures. We set all sky backgrounds to the "high" level in the ETC, and used the SUB2048 subarray with NRS readout pattern.

For 2M2139, we aimed for 3% precision and achieved  $\text{SNR} \sim 83$  with 50 groups/exposure for a cadence of 3 minutes. For PSO 318, we aimed for 1% precision and achieved  $\text{SNR} \sim 106$  with 100 groups/integration and 2 integrations/exposure, for a cadence of 12 minutes. These SNRs satisfied our requirements.

For 2M1126, a higher SNR is needed due to its lower J-band variability amplitude (1.2%). We settled for 0.25% precision or  $\text{SNR} \sim 400$ , or  $\sim 20$  minutes of exposure time (12 samples/3.5 hour period). This cadence should still allow for enough samplings in one rotation period. In order to minimize saturated pixels, we opt for 83 groups over 4 integrations for each 20-minute cadence.

# Proposal 8846 - Targets - The Polar Expedition: Extracting Spectra of Polar Regions in Planetary-mass Objects and Brown Dwarfs

| Fixed Targets | #                                                                                                                 | Name      | Target Coordinates                                                                   | Targ. Coord. Corrections                                                                                                        | Miscellaneous |
|---------------|-------------------------------------------------------------------------------------------------------------------|-----------|--------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|---------------|
|               | (1)                                                                                                               | 2M2139+02 | RA: 21 39 26.7694 (324.8615392d)<br>Dec: +02 20 22.70 (2.33964d)<br>Equinox: J2000   | Proper Motion RA: 485.9 mas/yr<br>Proper Motion Dec: 124.8 mas/yr<br>Parallax: 0.1015"<br>Epoch of Position: 2000               |               |
|               | Comments: SpT: T2, J-band variability: 26%, inclination: 90 degrees<br>Category=Star<br>Description=[T dwarfs]    |           |                                                                                      |                                                                                                                                 |               |
|               | (2)                                                                                                               | PSO318    | RA: 21 14 8.0256 (318.5334400d)<br>Dec: -22 51 35.84 (-22.85996d)<br>Equinox: J2000  | Proper Motion RA: 136.3 mas/yr<br>Proper Motion Dec: -144.29999994263198 mas/yr<br>Parallax: 0.0451"<br>Epoch of Position: 2000 |               |
|               | Comments: SpT: L7.5, J-band variability: 10%, inclination: 56 degrees<br>Category=Star<br>Description=[T dwarfs]  |           |                                                                                      |                                                                                                                                 |               |
|               | (3)                                                                                                               | 2M1126-50 | RA: 11 26 39.8047 (171.6658529d)<br>Dec: -50 03 54.85 (-50.06524d)<br>Equinox: J2000 | Proper Motion RA: -1589.907 mas/yr<br>Proper Motion Dec: 450.777 mas/yr<br>Parallax: 0.0618331"<br>Epoch of Position: 2000      |               |
|               | Comments: SpT: L6.5, J-band variability: 1.2%, inclination: 35 degrees<br>Category=Star<br>Description=[L dwarfs] |           |                                                                                      |                                                                                                                                 |               |

# Proposal 8846 - Observation 1 - The Polar Expedition: Extracting Spectra of Polar Regions in Planetary-mass Objects and Brown Dwa...

|                      |                                                                                                                                                                                                                     |                |                                                                                    |            |                  |                                                                                                                   |               |                    |                     |                     |                              |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|------------------------------------------------------------------------------------|------------|------------------|-------------------------------------------------------------------------------------------------------------------|---------------|--------------------|---------------------|---------------------|------------------------------|
| Observation          | Proposal 8846, Observation 1: 1A) 2M2139, ngroups=50, int/expo=303                                                                                                                                                  |                |                                                                                    |            |                  |                                                                                                                   |               |                    |                     |                     | Tue Mar 24 21:00:10 GMT 2026 |
|                      | Diagnostic Status: Warning                                                                                                                                                                                          |                |                                                                                    |            |                  |                                                                                                                   |               |                    |                     |                     |                              |
|                      | Observing Template: NIRSpec Bright Object Time Series                                                                                                                                                               |                |                                                                                    |            |                  |                                                                                                                   |               |                    |                     |                     |                              |
|                      | Comments: For 2M2139, we aimed for 3% precision and achieved SNR~83 with 50 groups/int, and 303 int/exposures.                                                                                                      |                |                                                                                    |            |                  |                                                                                                                   |               |                    |                     |                     |                              |
| Diagnostics          | (1A) 2M2139, ngroups=50, int/expo=303 (Obs 1)) Warning (Form): Exposure Duration exceeds the limit of 10000.0 seconds. Above this limit it is possible that a High Gain Antenna move may occur during the exposure. |                |                                                                                    |            |                  |                                                                                                                   |               |                    |                     |                     |                              |
|                      | (Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                                                         |                |                                                                                    |            |                  |                                                                                                                   |               |                    |                     |                     |                              |
| Fixed Targets        | #                                                                                                                                                                                                                   | Name           | Target Coordinates                                                                 |            |                  | Targ. Coord. Corrections                                                                                          |               |                    | Miscellaneous       |                     |                              |
|                      | (1)                                                                                                                                                                                                                 | 2M2139+02      | RA: 21 39 26.7694 (324.8615392d)<br>Dec: +02 20 22.70 (2.33964d)<br>Equinox: J2000 |            |                  | Proper Motion RA: 485.9 mas/yr<br>Proper Motion Dec: 124.8 mas/yr<br>Parallax: 0.1015"<br>Epoch of Position: 2000 |               |                    |                     |                     |                              |
|                      | Comments: SpT: T2, J-band variability: 26%, inclination: 90 degrees                                                                                                                                                 |                |                                                                                    |            |                  |                                                                                                                   |               |                    |                     |                     |                              |
|                      | Category=Star<br>Description=[T dwarfs]                                                                                                                                                                             |                |                                                                                    |            |                  |                                                                                                                   |               |                    |                     |                     |                              |
| Acquisition          | #                                                                                                                                                                                                                   | Target         | TA Method                                                                          | Subarray   | Filter           | Readout Pattern                                                                                                   | Groups/Int    | Integrations/Exp   | Total Integrations  | Total Exposure Time | Optional ETC ID              |
|                      | 1                                                                                                                                                                                                                   | SAME           | WATA                                                                               | SUB2048    | F110W            | NRSRAPIDD6                                                                                                        | 3             | 1                  | 1                   | 14.452              | 224186                       |
| Template             | Subarray                                                                                                                                                                                                            |                |                                                                                    |            |                  |                                                                                                                   |               |                    |                     |                     |                              |
|                      | SUB2048                                                                                                                                                                                                             |                |                                                                                    |            |                  |                                                                                                                   |               |                    |                     |                     |                              |
| Spectral Elements    | #                                                                                                                                                                                                                   | Grating/Filter | Readout Pattern                                                                    | Groups/Int | Integrations/Exp | Exposures/Dith                                                                                                    | Total Dithers | Total Integrations | Total Exposure Time | Optional ETC ID     |                              |
|                      | 1                                                                                                                                                                                                                   | G235H/F170LP   | NRS                                                                                | 50         | 303              | 1                                                                                                                 | 1             | 303                | 54940.711           | 224186              |                              |
| Special Requirements | Time Series Observation<br>No Parallel Attachments<br><br>4 After 1 by 114.3 Hours to <None specified>                                                                                                              |                |                                                                                    |            |                  |                                                                                                                   |               |                    |                     |                     |                              |

# Proposal 8846 - Observation 2 - The Polar Expedition: Extracting Spectra of Polar Regions in Planetary-mass Objects and Brown Dwa...

|                      |                                                                                                                                                                       |                |                                 |            |                  |                                               |               |                    |                     |                     |                              |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|---------------------------------|------------|------------------|-----------------------------------------------|---------------|--------------------|---------------------|---------------------|------------------------------|
| Observation          | Proposal 8846, Observation 2: 2A) PSO 318, ngroups=100x2, exposures/epoch=34                                                                                          |                |                                 |            |                  |                                               |               |                    |                     |                     | Tue Mar 24 21:00:10 GMT 2026 |
|                      | Diagnostic Status: Warning                                                                                                                                            |                |                                 |            |                  |                                               |               |                    |                     |                     |                              |
|                      | Observing Template: NIRSpec Bright Object Time Series                                                                                                                 |                |                                 |            |                  |                                               |               |                    |                     |                     |                              |
|                      | Comments: For PSO 318, we aimed for 1% precision and achieved SNR~106 with 100 groups/integration, 2 integrations/exposure, 34 exposures for a cadence of 12 minutes. |                |                                 |            |                  |                                               |               |                    |                     |                     |                              |
| Diagnostics          | (Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                           |                |                                 |            |                  |                                               |               |                    |                     |                     |                              |
| Fixed Targets        | #                                                                                                                                                                     | Name           | Target Coordinates              |            |                  | Targ. Coord. Corrections                      |               |                    | Miscellaneous       |                     |                              |
|                      | (2)                                                                                                                                                                   | PSO318         | RA: 21 14 8.0256 (318.5334400d) |            |                  | Proper Motion RA: 136.3 mas/yr                |               |                    |                     |                     |                              |
|                      |                                                                                                                                                                       |                | Dec: -22 51 35.84 (-22.85996d)  |            |                  | Proper Motion Dec: -144.29999994263198 mas/yr |               |                    |                     |                     |                              |
|                      |                                                                                                                                                                       |                | Equinox: J2000                  |            |                  | Parallax: 0.0451"                             |               |                    |                     |                     |                              |
|                      | Comments: SpT: L7.5, J-band variability: 10%, inclination: 56 degrees                                                                                                 |                |                                 |            |                  |                                               |               |                    |                     |                     |                              |
|                      | Category=Star                                                                                                                                                         |                |                                 |            |                  |                                               |               |                    |                     |                     |                              |
|                      | Description=[T dwarfs]                                                                                                                                                |                |                                 |            |                  |                                               |               |                    |                     |                     |                              |
| Acquisition          | #                                                                                                                                                                     | Target         | TA Method                       | Subarray   | Filter           | Readout Pattern                               | Groups/Int    | Integrations/Exp   | Total Integrations  | Total Exposure Time | Optional ETC ID              |
|                      | 1                                                                                                                                                                     | SAME           | WATA                            | SUB2048    | F110W            | NRSRAPIDD6                                    | 3             | 1                  | 1                   | 14.452              | 224186                       |
| Template             | Subarray                                                                                                                                                              |                |                                 |            |                  |                                               |               |                    |                     |                     |                              |
|                      | SUB2048                                                                                                                                                               |                |                                 |            |                  |                                               |               |                    |                     |                     |                              |
| Spectral Elements    | #                                                                                                                                                                     | Grating/Filter | Readout Pattern                 | Groups/Int | Integrations/Exp | Exposures/Dith                                | Total Dithers | Total Integrations | Total Exposure Time | Optional ETC ID     |                              |
|                      | 1                                                                                                                                                                     | G235H/F170LP   | NRS                             | 100        | 2                | 34                                            | 1             | 68                 | 24597.129           | 224186              |                              |
| Special Requirements | Time Series Observation<br>No Parallel Attachments<br><br>5 After 2 by 51 Hours to <None specified>                                                                   |                |                                 |            |                  |                                               |               |                    |                     |                     |                              |

# Proposal 8846 - Observation 3 - The Polar Expedition: Extracting Spectra of Polar Regions in Planetary-mass Objects and Brown Dwa...

|                      |                                                                                                                                                                                                                                                                                  |                |                                  |            |                  |                                    |               |                    |                     |                     |                              |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|----------------------------------|------------|------------------|------------------------------------|---------------|--------------------|---------------------|---------------------|------------------------------|
| Observation          | Proposal 8846, Observation 3: 3A) 2M1126, ngroups=83, int/expo=76                                                                                                                                                                                                                |                |                                  |            |                  |                                    |               |                    |                     |                     | Tue Mar 24 21:00:10 GMT 2026 |
|                      | Diagnostic Status: Warning                                                                                                                                                                                                                                                       |                |                                  |            |                  |                                    |               |                    |                     |                     |                              |
|                      | Observing Template: NIRSpec Bright Object Time Series                                                                                                                                                                                                                            |                |                                  |            |                  |                                    |               |                    |                     |                     |                              |
|                      | Comments: For 2M1126, a higher SNR is needed due to its lower J-band variability amplitude (1.2%). We settled for 0.4% precision or SNR~250, or ~5 minutes of exposure time (48 samples/3.5 hour period). This cadence should allow for enough samplings in one rotation period. |                |                                  |            |                  |                                    |               |                    |                     |                     |                              |
| Diagnostics          | (3A) 2M1126, ngroups=83, int/expo=76 (Obs 3)) Warning (Form): Exposure Duration exceeds the limit of 10000.0 seconds. Above this limit it is possible that a High Gain Antenna move may occur during the exposure.                                                               |                |                                  |            |                  |                                    |               |                    |                     |                     |                              |
|                      | (Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                                                                                                                      |                |                                  |            |                  |                                    |               |                    |                     |                     |                              |
| Fixed Targets        | #                                                                                                                                                                                                                                                                                | Name           | Target Coordinates               |            |                  | Targ. Coord. Corrections           |               |                    | Miscellaneous       |                     |                              |
|                      | (3)                                                                                                                                                                                                                                                                              | 2M1126-50      | RA: 11 26 39.8047 (171.6658529d) |            |                  | Proper Motion RA: -1589.907 mas/yr |               |                    |                     |                     |                              |
|                      |                                                                                                                                                                                                                                                                                  |                | Dec: -50 03 54.85 (-50.06524d)   |            |                  | Proper Motion Dec: 450.777 mas/yr  |               |                    |                     |                     |                              |
|                      |                                                                                                                                                                                                                                                                                  |                | Equinox: J2000                   |            |                  | Parallax: 0.0618331"               |               |                    |                     |                     |                              |
|                      | Epoch of Position: 2000                                                                                                                                                                                                                                                          |                |                                  |            |                  |                                    |               |                    |                     |                     |                              |
|                      | Comments: SpT: L6.5, J-band variability: 1.2%, inclination: 35 degrees                                                                                                                                                                                                           |                |                                  |            |                  |                                    |               |                    |                     |                     |                              |
|                      | Category=Star                                                                                                                                                                                                                                                                    |                |                                  |            |                  |                                    |               |                    |                     |                     |                              |
|                      | Description=[L dwarfs]                                                                                                                                                                                                                                                           |                |                                  |            |                  |                                    |               |                    |                     |                     |                              |
| Acquisition          | #                                                                                                                                                                                                                                                                                | Target         | TA Method                        | Subarray   | Filter           | Readout Pattern                    | Groups/Int    | Integrations/Exp   | Total Integrations  | Total Exposure Time | Optional ETC ID              |
|                      | 1                                                                                                                                                                                                                                                                                | SAME           | WATA                             | SUB2048    | F110W            | NRSRAPIDD6                         | 3             | 1                  | 1                   | 14.452              | 293646                       |
| Template             | Subarray                                                                                                                                                                                                                                                                         |                |                                  |            |                  |                                    |               |                    |                     |                     |                              |
|                      | SUB2048                                                                                                                                                                                                                                                                          |                |                                  |            |                  |                                    |               |                    |                     |                     |                              |
| Spectral Elements    | #                                                                                                                                                                                                                                                                                | Grating/Filter | Readout Pattern                  | Groups/Int | Integrations/Exp | Exposures/Dith                     | Total Dithers | Total Integrations | Total Exposure Time | Optional ETC ID     |                              |
|                      | 1                                                                                                                                                                                                                                                                                | G235H/F170LP   | NRS                              | 83         | 76               | 1                                  | 1             | 76                 | 22829.372           | 293646              |                              |
| Special Requirements | Time Series Observation                                                                                                                                                                                                                                                          |                |                                  |            |                  |                                    |               |                    |                     |                     |                              |
|                      | No Parallel Attachments                                                                                                                                                                                                                                                          |                |                                  |            |                  |                                    |               |                    |                     |                     |                              |
|                      | 6 After 3 by 48 Hours to <None specified>                                                                                                                                                                                                                                        |                |                                  |            |                  |                                    |               |                    |                     |                     |                              |

# Proposal 8846 - Observation 4 - The Polar Expedition: Extracting Spectra of Polar Regions in Planetary-mass Objects and Brown Dwa...

|                      |                                                                                                                                                                                                                     |                |                                                                                    |            |                  |                                                                                                                   |               |                    |                     |                     |                              |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|------------------------------------------------------------------------------------|------------|------------------|-------------------------------------------------------------------------------------------------------------------|---------------|--------------------|---------------------|---------------------|------------------------------|
| Observation          | Proposal 8846, Observation 4: 1B) 2M2139, ngroups=50, int/expo=303                                                                                                                                                  |                |                                                                                    |            |                  |                                                                                                                   |               |                    |                     |                     | Tue Mar 24 21:00:10 GMT 2026 |
|                      | Diagnostic Status: Warning                                                                                                                                                                                          |                |                                                                                    |            |                  |                                                                                                                   |               |                    |                     |                     |                              |
|                      | Observing Template: NIRSpec Bright Object Time Series                                                                                                                                                               |                |                                                                                    |            |                  |                                                                                                                   |               |                    |                     |                     |                              |
|                      | Comments: For 2M2139, we aimed for 3% precision and achieved SNR~83 with 50 groups/int, and 303 int/exposures.                                                                                                      |                |                                                                                    |            |                  |                                                                                                                   |               |                    |                     |                     |                              |
| Diagnostics          | (1B) 2M2139, ngroups=50, int/expo=303 (Obs 4)) Warning (Form): Exposure Duration exceeds the limit of 10000.0 seconds. Above this limit it is possible that a High Gain Antenna move may occur during the exposure. |                |                                                                                    |            |                  |                                                                                                                   |               |                    |                     |                     |                              |
|                      | (Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                                                         |                |                                                                                    |            |                  |                                                                                                                   |               |                    |                     |                     |                              |
| Fixed Targets        | #                                                                                                                                                                                                                   | Name           | Target Coordinates                                                                 |            |                  | Targ. Coord. Corrections                                                                                          |               |                    | Miscellaneous       |                     |                              |
|                      | (1)                                                                                                                                                                                                                 | 2M2139+02      | RA: 21 39 26.7694 (324.8615392d)<br>Dec: +02 20 22.70 (2.33964d)<br>Equinox: J2000 |            |                  | Proper Motion RA: 485.9 mas/yr<br>Proper Motion Dec: 124.8 mas/yr<br>Parallax: 0.1015"<br>Epoch of Position: 2000 |               |                    |                     |                     |                              |
|                      | Comments: SpT: T2, J-band variability: 26%, inclination: 90 degrees                                                                                                                                                 |                |                                                                                    |            |                  |                                                                                                                   |               |                    |                     |                     |                              |
|                      | Category=Star<br>Description=[T dwarfs]                                                                                                                                                                             |                |                                                                                    |            |                  |                                                                                                                   |               |                    |                     |                     |                              |
| Acquisition          | #                                                                                                                                                                                                                   | Target         | TA Method                                                                          | Subarray   | Filter           | Readout Pattern                                                                                                   | Groups/Int    | Integrations/Exp   | Total Integrations  | Total Exposure Time | Optional ETC ID              |
|                      | 1                                                                                                                                                                                                                   | SAME           | WATA                                                                               | SUB2048    | F110W            | NRSRAPIDD6                                                                                                        | 3             | 1                  | 1                   | 14.452              | 224186                       |
| Template             | Subarray                                                                                                                                                                                                            |                |                                                                                    |            |                  |                                                                                                                   |               |                    |                     |                     |                              |
|                      | SUB2048                                                                                                                                                                                                             |                |                                                                                    |            |                  |                                                                                                                   |               |                    |                     |                     |                              |
| Spectral Elements    | #                                                                                                                                                                                                                   | Grating/Filter | Readout Pattern                                                                    | Groups/Int | Integrations/Exp | Exposures/Dith                                                                                                    | Total Dithers | Total Integrations | Total Exposure Time | Optional ETC ID     |                              |
|                      | 1                                                                                                                                                                                                                   | G235H/F170LP   | NRS                                                                                | 50         | 303              | 1                                                                                                                 | 1             | 303                | 54940.711           | 224186              |                              |
| Special Requirements | Time Series Observation<br>No Parallel Attachments<br><br>4 After 1 by 114.3 Hours to <None specified><br>7 After 4 by 114.3 Hours to <None specified>                                                              |                |                                                                                    |            |                  |                                                                                                                   |               |                    |                     |                     |                              |



# Proposal 8846 - Observation 5 - The Polar Expedition: Extracting Spectra of Polar Regions in Planetary-mass Objects and Brown Dwa...

|                      |                                                                                                                                                                       |                |                                 |            |                  |                                               |               |                    |                     |                     |                              |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|---------------------------------|------------|------------------|-----------------------------------------------|---------------|--------------------|---------------------|---------------------|------------------------------|
| Observation          | Proposal 8846, Observation 5: 2B) PSO 318, ngroups=100x2, exposures/epoch=34                                                                                          |                |                                 |            |                  |                                               |               |                    |                     |                     | Tue Mar 24 21:00:10 GMT 2026 |
|                      | Diagnostic Status: Warning                                                                                                                                            |                |                                 |            |                  |                                               |               |                    |                     |                     |                              |
|                      | Observing Template: NIRSpec Bright Object Time Series                                                                                                                 |                |                                 |            |                  |                                               |               |                    |                     |                     |                              |
|                      | Comments: For PSO 318, we aimed for 1% precision and achieved SNR~106 with 100 groups/integration, 2 integrations/exposure, 34 exposures for a cadence of 12 minutes. |                |                                 |            |                  |                                               |               |                    |                     |                     |                              |
| Diagnostics          | (Visit 5:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                           |                |                                 |            |                  |                                               |               |                    |                     |                     |                              |
| Fixed Targets        | #                                                                                                                                                                     | Name           | Target Coordinates              |            |                  | Targ. Coord. Corrections                      |               |                    | Miscellaneous       |                     |                              |
|                      | (2)                                                                                                                                                                   | PSO318         | RA: 21 14 8.0256 (318.5334400d) |            |                  | Proper Motion RA: 136.3 mas/yr                |               |                    |                     |                     |                              |
|                      |                                                                                                                                                                       |                | Dec: -22 51 35.84 (-22.85996d)  |            |                  | Proper Motion Dec: -144.29999994263198 mas/yr |               |                    |                     |                     |                              |
|                      |                                                                                                                                                                       |                | Equinox: J2000                  |            |                  | Parallax: 0.0451"                             |               |                    |                     |                     |                              |
|                      | Comments: SpT: L7.5, J-band variability: 10%, inclination: 56 degrees                                                                                                 |                |                                 |            |                  |                                               |               |                    |                     |                     |                              |
|                      | Category=Star                                                                                                                                                         |                |                                 |            |                  |                                               |               |                    |                     |                     |                              |
|                      | Description=[T dwarfs]                                                                                                                                                |                |                                 |            |                  |                                               |               |                    |                     |                     |                              |
| Acquisition          | #                                                                                                                                                                     | Target         | TA Method                       | Subarray   | Filter           | Readout Pattern                               | Groups/Int    | Integrations/Exp   | Total Integrations  | Total Exposure Time | Optional ETC ID              |
|                      | 1                                                                                                                                                                     | SAME           | WATA                            | SUB2048    | F110W            | NRSRAPIDD6                                    | 3             | 1                  | 1                   | 14.452              | 224186                       |
| Template             | Subarray                                                                                                                                                              |                |                                 |            |                  |                                               |               |                    |                     |                     |                              |
|                      | SUB2048                                                                                                                                                               |                |                                 |            |                  |                                               |               |                    |                     |                     |                              |
| Spectral Elements    | #                                                                                                                                                                     | Grating/Filter | Readout Pattern                 | Groups/Int | Integrations/Exp | Exposures/Dith                                | Total Dithers | Total Integrations | Total Exposure Time | Optional ETC ID     |                              |
|                      | 1                                                                                                                                                                     | G235H/F170LP   | NRS                             | 100        | 2                | 34                                            | 1             | 68                 | 24597.129           | 224186              |                              |
| Special Requirements | Time Series Observation<br>No Parallel Attachments<br><br>5 After 2 by 51 Hours to <None specified><br>8 After 5 by 51 Hours to <None specified>                      |                |                                 |            |                  |                                               |               |                    |                     |                     |                              |

# Proposal 8846 - Observation 6 - The Polar Expedition: Extracting Spectra of Polar Regions in Planetary-mass Objects and Brown Dwa...

|                                                                        |                                                                                                                                                                                                                                                                                  |                |                                  |                                   |                  |                                    |               |                    |                     |                     |                              |
|------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|----------------------------------|-----------------------------------|------------------|------------------------------------|---------------|--------------------|---------------------|---------------------|------------------------------|
| Observation                                                            | Proposal 8846, Observation 6: 3B) 2M1126, ngroups=83, int/expo=76                                                                                                                                                                                                                |                |                                  |                                   |                  |                                    |               |                    |                     |                     | Tue Mar 24 21:00:10 GMT 2026 |
|                                                                        | Diagnostic Status: Warning                                                                                                                                                                                                                                                       |                |                                  |                                   |                  |                                    |               |                    |                     |                     |                              |
|                                                                        | Observing Template: NIRSpec Bright Object Time Series                                                                                                                                                                                                                            |                |                                  |                                   |                  |                                    |               |                    |                     |                     |                              |
|                                                                        | Comments: For 2M1126, a higher SNR is needed due to its lower J-band variability amplitude (1.2%). We settled for 0.4% precision or SNR~250, or ~5 minutes of exposure time (48 samples/3.5 hour period). This cadence should allow for enough samplings in one rotation period. |                |                                  |                                   |                  |                                    |               |                    |                     |                     |                              |
| Diagnostics                                                            | (3B) 2M1126, ngroups=83, int/expo=76 (Obs 6)) Warning (Form): Exposure Duration exceeds the limit of 10000.0 seconds. Above this limit it is possible that a High Gain Antenna move may occur during the exposure.                                                               |                |                                  |                                   |                  |                                    |               |                    |                     |                     |                              |
|                                                                        | (Visit 6:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                                                                                                                      |                |                                  |                                   |                  |                                    |               |                    |                     |                     |                              |
| Fixed Targets                                                          | #                                                                                                                                                                                                                                                                                | Name           | Target Coordinates               |                                   |                  | Targ. Coord. Corrections           |               |                    | Miscellaneous       |                     |                              |
|                                                                        | (3)                                                                                                                                                                                                                                                                              | 2M1126-50      | RA: 11 26 39.8047 (171.6658529d) |                                   |                  | Proper Motion RA: -1589.907 mas/yr |               |                    |                     |                     |                              |
|                                                                        | Dec: -50 03 54.85 (-50.06524d)                                                                                                                                                                                                                                                   |                |                                  | Proper Motion Dec: 450.777 mas/yr |                  |                                    |               |                    |                     |                     |                              |
|                                                                        | Equinox: J2000                                                                                                                                                                                                                                                                   |                |                                  | Parallax: 0.0618331"              |                  |                                    |               |                    |                     |                     |                              |
| Epoch of Position: 2000                                                |                                                                                                                                                                                                                                                                                  |                |                                  |                                   |                  |                                    |               |                    |                     |                     |                              |
| Comments: SpT: L6.5, J-band variability: 1.2%, inclination: 35 degrees |                                                                                                                                                                                                                                                                                  |                |                                  |                                   |                  |                                    |               |                    |                     |                     |                              |
| Category=Star                                                          |                                                                                                                                                                                                                                                                                  |                |                                  |                                   |                  |                                    |               |                    |                     |                     |                              |
| Description=[L dwarfs]                                                 |                                                                                                                                                                                                                                                                                  |                |                                  |                                   |                  |                                    |               |                    |                     |                     |                              |
| Acquisition                                                            | #                                                                                                                                                                                                                                                                                | Target         | TA Method                        | Subarray                          | Filter           | Readout Pattern                    | Groups/Int    | Integrations/Exp   | Total Integrations  | Total Exposure Time | Optional ETC ID              |
|                                                                        | 1                                                                                                                                                                                                                                                                                | SAME           | WATA                             | SUB2048                           | F110W            | NRSRAPIDD6                         | 3             | 1                  | 1                   | 14.452              | 224186                       |
| Template                                                               | Subarray                                                                                                                                                                                                                                                                         |                |                                  |                                   |                  |                                    |               |                    |                     |                     |                              |
|                                                                        | SUB2048                                                                                                                                                                                                                                                                          |                |                                  |                                   |                  |                                    |               |                    |                     |                     |                              |
| Spectral Elements                                                      | #                                                                                                                                                                                                                                                                                | Grating/Filter | Readout Pattern                  | Groups/Int                        | Integrations/Exp | Exposures/Dith                     | Total Dithers | Total Integrations | Total Exposure Time | Optional ETC ID     |                              |
|                                                                        | 1                                                                                                                                                                                                                                                                                | G235H/F170LP   | NRS                              | 83                                | 76               | 1                                  | 1             | 76                 | 22829.372           | 224186              |                              |
| Special Requirements                                                   | Time Series Observation                                                                                                                                                                                                                                                          |                |                                  |                                   |                  |                                    |               |                    |                     |                     |                              |
|                                                                        | No Parallel Attachments                                                                                                                                                                                                                                                          |                |                                  |                                   |                  |                                    |               |                    |                     |                     |                              |
| 6 After 3 by 48 Hours to <None specified>                              |                                                                                                                                                                                                                                                                                  |                |                                  |                                   |                  |                                    |               |                    |                     |                     |                              |
| 9 After 6 by 48 Hours to <None specified>                              |                                                                                                                                                                                                                                                                                  |                |                                  |                                   |                  |                                    |               |                    |                     |                     |                              |

# Proposal 8846 - Observation 7 - The Polar Expedition: Extracting Spectra of Polar Regions in Planetary-mass Objects and Brown Dwa...

|                      |                                                                                                                                                                                                                     |                |                                                                                    |            |                  |                                                                                                                   |               |                    |                     |                     |                              |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|------------------------------------------------------------------------------------|------------|------------------|-------------------------------------------------------------------------------------------------------------------|---------------|--------------------|---------------------|---------------------|------------------------------|
| Observation          | Proposal 8846, Observation 7: 1C) 2M2139, ngroups=50, int/expo=303                                                                                                                                                  |                |                                                                                    |            |                  |                                                                                                                   |               |                    |                     |                     | Tue Mar 24 21:00:10 GMT 2026 |
|                      | Diagnostic Status: Warning                                                                                                                                                                                          |                |                                                                                    |            |                  |                                                                                                                   |               |                    |                     |                     |                              |
|                      | Observing Template: NIRSpec Bright Object Time Series                                                                                                                                                               |                |                                                                                    |            |                  |                                                                                                                   |               |                    |                     |                     |                              |
|                      | Comments: For 2M2139, we aimed for 3% precision and achieved SNR~83 with 50 groups/int, and 303 int/exposures.                                                                                                      |                |                                                                                    |            |                  |                                                                                                                   |               |                    |                     |                     |                              |
| Diagnostics          | (1C) 2M2139, ngroups=50, int/expo=303 (Obs 7)) Warning (Form): Exposure Duration exceeds the limit of 10000.0 seconds. Above this limit it is possible that a High Gain Antenna move may occur during the exposure. |                |                                                                                    |            |                  |                                                                                                                   |               |                    |                     |                     |                              |
|                      | (Visit 7:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                                                         |                |                                                                                    |            |                  |                                                                                                                   |               |                    |                     |                     |                              |
| Fixed Targets        | #                                                                                                                                                                                                                   | Name           | Target Coordinates                                                                 |            |                  | Targ. Coord. Corrections                                                                                          |               |                    | Miscellaneous       |                     |                              |
|                      | (1)                                                                                                                                                                                                                 | 2M2139+02      | RA: 21 39 26.7694 (324.8615392d)<br>Dec: +02 20 22.70 (2.33964d)<br>Equinox: J2000 |            |                  | Proper Motion RA: 485.9 mas/yr<br>Proper Motion Dec: 124.8 mas/yr<br>Parallax: 0.1015"<br>Epoch of Position: 2000 |               |                    |                     |                     |                              |
|                      | Comments: SpT: T2, J-band variability: 26%, inclination: 90 degrees                                                                                                                                                 |                |                                                                                    |            |                  |                                                                                                                   |               |                    |                     |                     |                              |
|                      | Category=Star<br>Description=[T dwarfs]                                                                                                                                                                             |                |                                                                                    |            |                  |                                                                                                                   |               |                    |                     |                     |                              |
| Acquisition          | #                                                                                                                                                                                                                   | Target         | TA Method                                                                          | Subarray   | Filter           | Readout Pattern                                                                                                   | Groups/Int    | Integrations/Exp   | Total Integrations  | Total Exposure Time | Optional ETC ID              |
|                      | 1                                                                                                                                                                                                                   | SAME           | WATA                                                                               | SUB2048    | F110W            | NRSRAPIDD6                                                                                                        | 3             | 1                  | 1                   | 14.452              | 224186                       |
| Template             | Subarray                                                                                                                                                                                                            |                |                                                                                    |            |                  |                                                                                                                   |               |                    |                     |                     |                              |
|                      | SUB2048                                                                                                                                                                                                             |                |                                                                                    |            |                  |                                                                                                                   |               |                    |                     |                     |                              |
| Spectral Elements    | #                                                                                                                                                                                                                   | Grating/Filter | Readout Pattern                                                                    | Groups/Int | Integrations/Exp | Exposures/Dith                                                                                                    | Total Dithers | Total Integrations | Total Exposure Time | Optional ETC ID     |                              |
|                      | 1                                                                                                                                                                                                                   | G235H/F170LP   | NRS                                                                                | 50         | 303              | 1                                                                                                                 | 1             | 303                | 54940.711           | 224186              |                              |
| Special Requirements | Time Series Observation<br>No Parallel Attachments<br><br>7 After 4 by 114.3 Hours to <None specified>                                                                                                              |                |                                                                                    |            |                  |                                                                                                                   |               |                    |                     |                     |                              |

# Proposal 8846 - Observation 8 - The Polar Expedition: Extracting Spectra of Polar Regions in Planetary-mass Objects and Brown Dwa...

|                      |                                                                                                                                                                       |                |                                                                                     |            |                  |                                                                                                                                 |               |                    |                     |                     |                              |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|-------------------------------------------------------------------------------------|------------|------------------|---------------------------------------------------------------------------------------------------------------------------------|---------------|--------------------|---------------------|---------------------|------------------------------|
| Observation          | Proposal 8846, Observation 8: 2C) PSO 318, ngroups=100x2, exposures/epoch=34                                                                                          |                |                                                                                     |            |                  |                                                                                                                                 |               |                    |                     |                     | Tue Mar 24 21:00:10 GMT 2026 |
|                      | Diagnostic Status: Warning                                                                                                                                            |                |                                                                                     |            |                  |                                                                                                                                 |               |                    |                     |                     |                              |
|                      | Observing Template: NIRSpec Bright Object Time Series                                                                                                                 |                |                                                                                     |            |                  |                                                                                                                                 |               |                    |                     |                     |                              |
|                      | Comments: For PSO 318, we aimed for 1% precision and achieved SNR~106 with 100 groups/integration, 2 integrations/exposure, 34 exposures for a cadence of 12 minutes. |                |                                                                                     |            |                  |                                                                                                                                 |               |                    |                     |                     |                              |
| Diagnostics          | (Visit 8:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                           |                |                                                                                     |            |                  |                                                                                                                                 |               |                    |                     |                     |                              |
| Fixed Targets        | #                                                                                                                                                                     | Name           | Target Coordinates                                                                  |            |                  | Targ. Coord. Corrections                                                                                                        |               |                    | Miscellaneous       |                     |                              |
|                      | (2)                                                                                                                                                                   | PSO318         | RA: 21 14 8.0256 (318.5334400d)<br>Dec: -22 51 35.84 (-22.85996d)<br>Equinox: J2000 |            |                  | Proper Motion RA: 136.3 mas/yr<br>Proper Motion Dec: -144.29999994263198 mas/yr<br>Parallax: 0.0451"<br>Epoch of Position: 2000 |               |                    |                     |                     |                              |
|                      | Comments: SpT: L7.5, J-band variability: 10%, inclination: 56 degrees                                                                                                 |                |                                                                                     |            |                  |                                                                                                                                 |               |                    |                     |                     |                              |
|                      | Category=Star<br>Description=[T dwarfs]                                                                                                                               |                |                                                                                     |            |                  |                                                                                                                                 |               |                    |                     |                     |                              |
| Acquisition          | #                                                                                                                                                                     | Target         | TA Method                                                                           | Subarray   | Filter           | Readout Pattern                                                                                                                 | Groups/Int    | Integrations/Exp   | Total Integrations  | Total Exposure Time | Optional ETC ID              |
|                      | 1                                                                                                                                                                     | SAME           | WATA                                                                                | SUB2048    | F110W            | NRSRAPIDD6                                                                                                                      | 3             | 1                  | 1                   | 14.452              | 224186                       |
| Template             | Subarray                                                                                                                                                              |                |                                                                                     |            |                  |                                                                                                                                 |               |                    |                     |                     |                              |
|                      | SUB2048                                                                                                                                                               |                |                                                                                     |            |                  |                                                                                                                                 |               |                    |                     |                     |                              |
| Spectral Elements    | #                                                                                                                                                                     | Grating/Filter | Readout Pattern                                                                     | Groups/Int | Integrations/Exp | Exposures/Dith                                                                                                                  | Total Dithers | Total Integrations | Total Exposure Time | Optional ETC ID     |                              |
|                      | 1                                                                                                                                                                     | G235H/F170LP   | NRS                                                                                 | 100        | 2                | 34                                                                                                                              | 1             | 68                 | 24597.129           | 224186              |                              |
| Special Requirements | Time Series Observation<br>No Parallel Attachments<br><br>8 After 5 by 51 Hours to <None specified>                                                                   |                |                                                                                     |            |                  |                                                                                                                                 |               |                    |                     |                     |                              |

# Proposal 8846 - Observation 9 - The Polar Expedition: Extracting Spectra of Polar Regions in Planetary-mass Objects and Brown Dwa...

|                                                                        |                                                                                                                                                                                                                                                                                  |                |                                  |                                   |                  |                                    |               |                    |                     |                     |                              |
|------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|----------------------------------|-----------------------------------|------------------|------------------------------------|---------------|--------------------|---------------------|---------------------|------------------------------|
| Observation                                                            | Proposal 8846, Observation 9: 3C) 2M1126, ngroups=83, int/expo=76                                                                                                                                                                                                                |                |                                  |                                   |                  |                                    |               |                    |                     |                     | Tue Mar 24 21:00:10 GMT 2026 |
|                                                                        | Diagnostic Status: Warning                                                                                                                                                                                                                                                       |                |                                  |                                   |                  |                                    |               |                    |                     |                     |                              |
|                                                                        | Observing Template: NIRSpec Bright Object Time Series                                                                                                                                                                                                                            |                |                                  |                                   |                  |                                    |               |                    |                     |                     |                              |
|                                                                        | Comments: For 2M1126, a higher SNR is needed due to its lower J-band variability amplitude (1.2%). We settled for 0.4% precision or SNR~250, or ~5 minutes of exposure time (48 samples/3.5 hour period). This cadence should allow for enough samplings in one rotation period. |                |                                  |                                   |                  |                                    |               |                    |                     |                     |                              |
| Diagnostics                                                            | (3C) 2M1126, ngroups=83, int/expo=76 (Obs 9)) Warning (Form): Exposure Duration exceeds the limit of 10000.0 seconds. Above this limit it is possible that a High Gain Antenna move may occur during the exposure.                                                               |                |                                  |                                   |                  |                                    |               |                    |                     |                     |                              |
|                                                                        | (Visit 9:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                                                                                                                      |                |                                  |                                   |                  |                                    |               |                    |                     |                     |                              |
| Fixed Targets                                                          | #                                                                                                                                                                                                                                                                                | Name           | Target Coordinates               |                                   |                  | Targ. Coord. Corrections           |               |                    | Miscellaneous       |                     |                              |
|                                                                        | (3)                                                                                                                                                                                                                                                                              | 2M1126-50      | RA: 11 26 39.8047 (171.6658529d) |                                   |                  | Proper Motion RA: -1589.907 mas/yr |               |                    |                     |                     |                              |
|                                                                        | Dec: -50 03 54.85 (-50.06524d)                                                                                                                                                                                                                                                   |                |                                  | Proper Motion Dec: 450.777 mas/yr |                  |                                    |               |                    |                     |                     |                              |
|                                                                        | Equinox: J2000                                                                                                                                                                                                                                                                   |                |                                  | Parallax: 0.0618331"              |                  |                                    |               |                    |                     |                     |                              |
|                                                                        |                                                                                                                                                                                                                                                                                  |                | Epoch of Position: 2000          |                                   |                  |                                    |               |                    |                     |                     |                              |
| Comments: SpT: L6.5, J-band variability: 1.2%, inclination: 35 degrees |                                                                                                                                                                                                                                                                                  |                |                                  |                                   |                  |                                    |               |                    |                     |                     |                              |
| Category=Star                                                          |                                                                                                                                                                                                                                                                                  |                |                                  |                                   |                  |                                    |               |                    |                     |                     |                              |
| Description=[L dwarfs]                                                 |                                                                                                                                                                                                                                                                                  |                |                                  |                                   |                  |                                    |               |                    |                     |                     |                              |
| Acquisition                                                            | #                                                                                                                                                                                                                                                                                | Target         | TA Method                        | Subarray                          | Filter           | Readout Pattern                    | Groups/Int    | Integrations/Exp   | Total Integrations  | Total Exposure Time | Optional ETC ID              |
|                                                                        | 1                                                                                                                                                                                                                                                                                | SAME           | WATA                             | SUB2048                           | F110W            | NRSRAPIDD6                         | 3             | 1                  | 1                   | 14.452              | 224186                       |
| Template                                                               | Subarray                                                                                                                                                                                                                                                                         |                |                                  |                                   |                  |                                    |               |                    |                     |                     |                              |
|                                                                        | SUB2048                                                                                                                                                                                                                                                                          |                |                                  |                                   |                  |                                    |               |                    |                     |                     |                              |
| Spectral Elements                                                      | #                                                                                                                                                                                                                                                                                | Grating/Filter | Readout Pattern                  | Groups/Int                        | Integrations/Exp | Exposures/Dith                     | Total Dithers | Total Integrations | Total Exposure Time | Optional ETC ID     |                              |
|                                                                        | 1                                                                                                                                                                                                                                                                                | G235H/F170LP   | NRS                              | 83                                | 76               | 1                                  | 1             | 76                 | 22829.372           | 224186              |                              |
| Special Requirements                                                   | Time Series Observation                                                                                                                                                                                                                                                          |                |                                  |                                   |                  |                                    |               |                    |                     |                     |                              |
|                                                                        | No Parallel Attachments                                                                                                                                                                                                                                                          |                |                                  |                                   |                  |                                    |               |                    |                     |                     |                              |
|                                                                        | 9 After 6 by 48 Hours to <None specified>                                                                                                                                                                                                                                        |                |                                  |                                   |                  |                                    |               |                    |                     |                     |                              |